CRL-LS-CRPA4 - L1 + L2/L5 + NavIC S 4-channel CRPA front end

Design report, Rev A

BEL - CRL, Antenna / RFM lab. Date 06-11-2026. Project folder `projects/02_ls-crpa4-rf-module`.

1. Summary

CRL-LS-CRPA4 is a D200 mm, 4-layer active antenna board for a 4-element controlled-reception-pattern antenna (CRPA). Each element has a stacked GPS L1/L2 patch (GPDF6010.A) and a NavIC S patch (CRL-LS-SP01, BEL CRL). Every element feeds three filtered, amplified branches (L1, L2/L5, S): 2 LNAs and 3 SAW filters per branch, as in projects 16 and 17. L1 and L2/L5 are combined in a DXP.02.A diplexer per channel. Outputs: 4 L SMA + 4 S SMA, +5 V on one DC connector. A machined aluminium back cover (BC01) gives every chain its own shielded cavity.

Case	Gain dB	NF dB	Input P1dB dBm	IIP3 dBm	2nd SAW rating dBm	Antenna CW that reaches it dBm
GPS L1 C/A (1575.42 MHz)	37.1	1.01	-20.4	-7.9	15	-27.4
GPS L5 (1176.45 MHz)	40.8	0.79	-23.9	-11.4	5	-40.9
GPS L2C (1227.60 MHz)	38.3	0.83	-22.9	-10.4	5	-39.9
NavIC S (2492.028 MHz)	33.0	1.35	-15.2	-2.2	10	-26.1

Supply 1.68 A typ / 2.16 A max at 5 V (8.4 W typ). Board DRC: 0 violations, 0 unconnected. Fabrication package complete (Gerber X2, Excellon, pick-and-place, IPC-D-356, BOM with 40 lines / 313 placements).

Key findings (section 7):

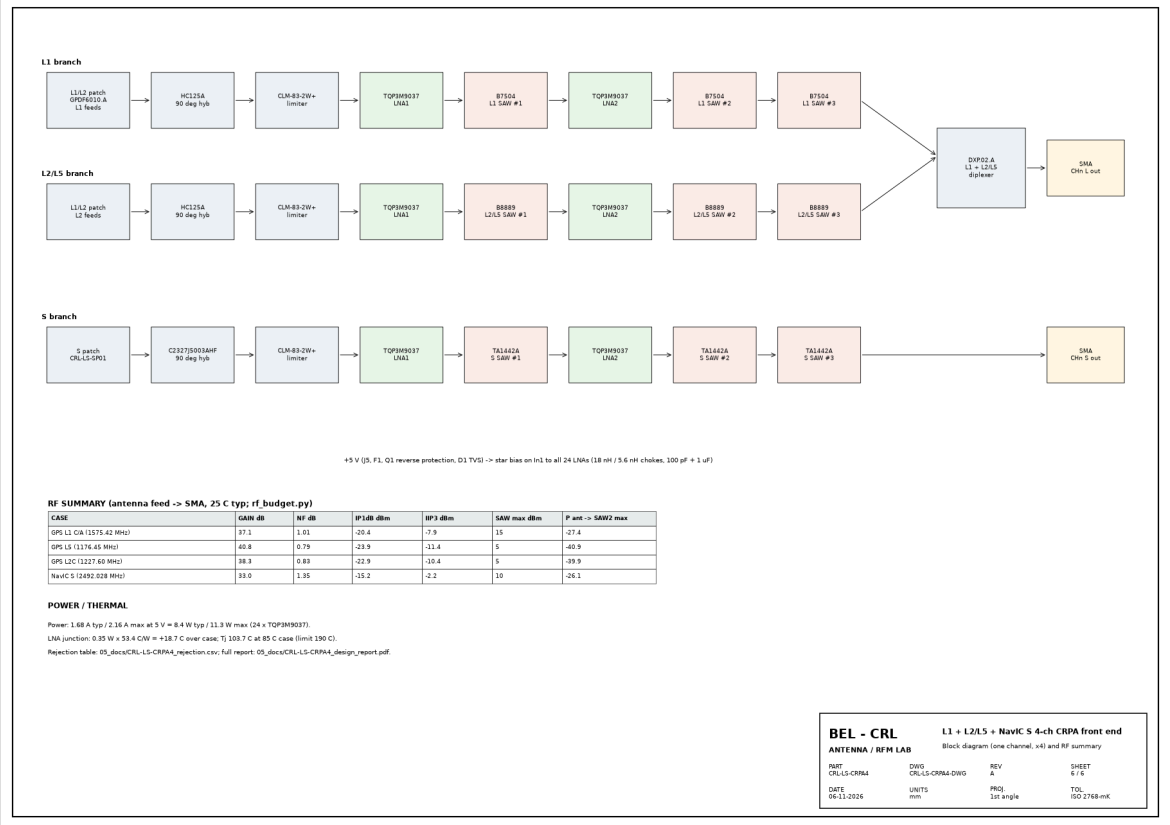
1. Weak rejection close to the bands: DME / TACAN 1150 9.9 dB; WiFi ch11 2462 4.4 dB; WiFi ch13 / BT high 2472-2480 2.2 dB; LTE B41 low 2506 -0.1 dB; LTE B7 UL 2535 5.9 dB.
2. Gain and filter ratings: an in-band CW jammer of -41 dBm (L2/L5), -27 dBm (L1) or -26 dBm (S) at the antenna drives the 2nd SAW to its power rating. The input 1 dB compression point is -24 to -20 dBm (L) and -15 dBm (S). For a CRPA, which must stay linear under jamming, this is the main Rev B item.

2. Design basis and decisions

Item	Decision	Source
Bands	GPS L1 1575.42; wide L2/L5 1164-1240; NavIC S 2492.028 MHz	user, 2026-11-05
Elements	4, square, sequential rotation (-90 deg x k)	customer board combine_model_2.0

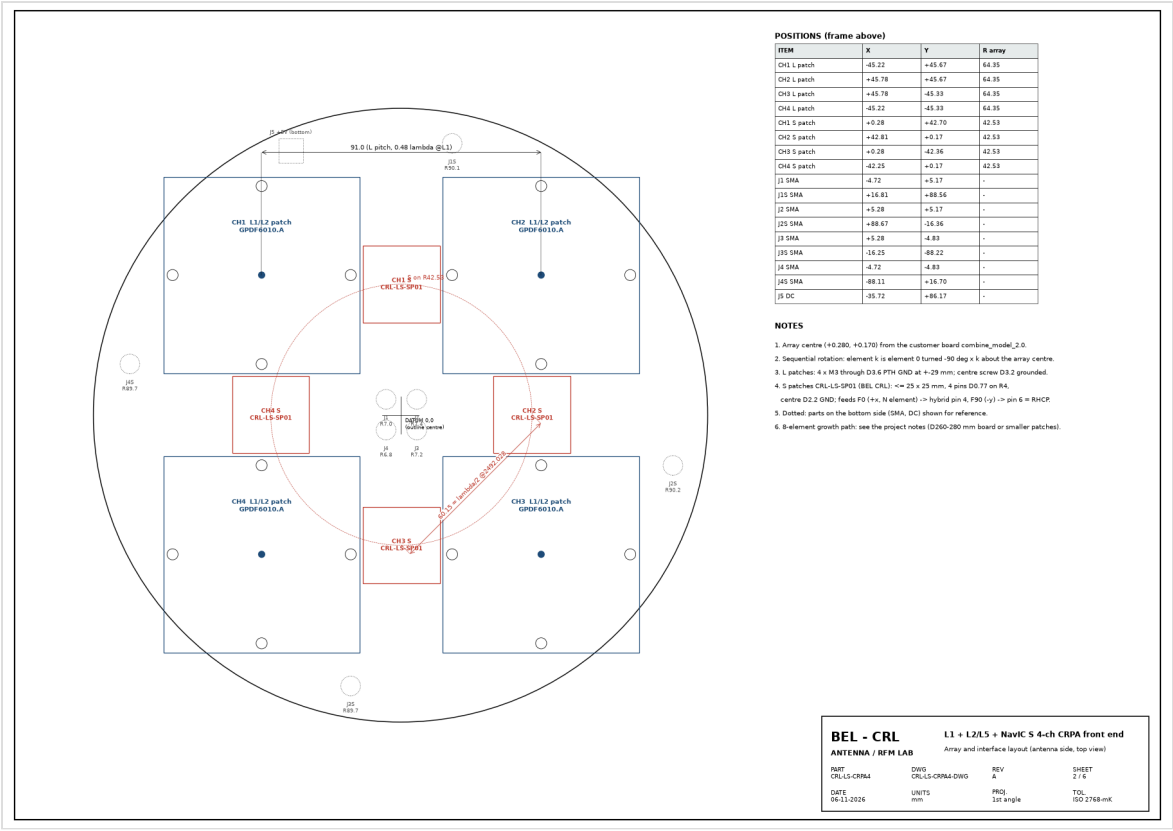
Item	Decision	Source
L spacing	91.0 mm (0.48 lambda at L1)	customer board
S spacing	60.15 mm = lambda/2 at 2492.028 MHz, R 42.53 mm	user, 2026-11-06
Chain	limiter -> LNA -> SAW -> LNA -> SAW -> SAW per branch	projects 16 / 17
Supply	+5 V, one DC connector, reverse-polarity PFET, fuse, TVS	user
Outputs	8 SMA (L and S separate) + DC	user
Board	D200 mm, same size as the L-band board	user

3. Architecture



Per element: the patch feeds go to a 90 deg hybrid (HC125A for L1 and L2, Xinger C2327J5003AHF for S), which combines them into RHCP. The hybrid's isolated port goes to 50 ohm. Next comes a CLM-83-2W+ PIN limiter (CW +32 dBm, limited output +11.5 dBm typ), then the LNA / SAW cascade. S hybrid wiring: F0 -> pin 4 (direct), F90 -> pin 6 (coupled). This is RHCP because F90 sits +90 deg counter-clockwise (top view) from F0. `build_pcb.py` checks it on every build.

Array geometry



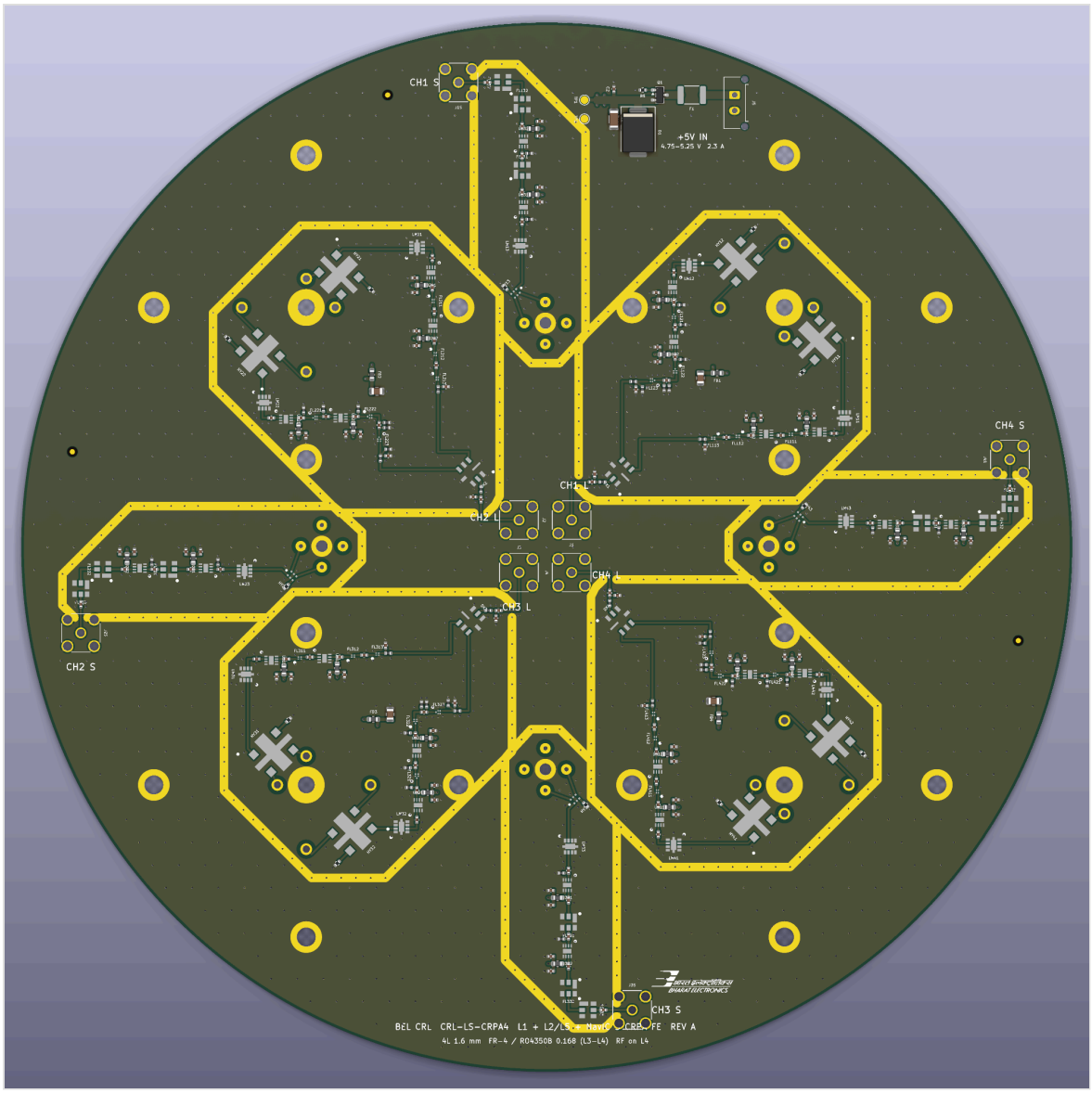
The S patches sit at lambda/2 (60.15 mm) so the S sub-array has no grating lobes and keeps full nulling depth. Each S patch clears its L neighbours by about 1 mm (25 x 25 mm patch). Mutual coupling L-to-S must be measured (section 9).

4. Stack-up and transmission lines

Layer	Type	mm	er	Note
F.Mask	mask	0.010	3.8	green LPI
F.Cu	copper	0.035		L1: solid GND = antenna ground plane
Core1	core	0.710	4.4	FR-4 high-Tg core (IT-180A)
In1.Cu	copper	0.035		L2: GND + per-channel +5V star traces
PP1	prepreg	0.540	4.3	FR-4 prepreg 2x2116 + 1x1080 (bonds FR-4 to RO4350B)
In2.Cu	copper	0.035		L3: solid GND, RF reference for L4
Core2	core	0.168	3.66	Rogers RO4350B 0.168 mm (6.6 mil), tan d 0.0037
B.Cu	copper	0.035		L4: RF lines + all SMD parts, ENIG
B.Mask	mask	0.010	3.8	green LPI

The 50 ohm microstrip on L4 is 0.34 mm over the RO4350B 0.168 mm core to the L3 ground. Loss is 0.066 dB/cm at 2.5 GHz, against 0.121 dB/cm for the same line on FR-4 7628. L1 is the solid antenna ground plane. L2 carries the per-channel +5 V star traces inside GND. Fence vias run along every RF line; frame vias sit at 2.5 mm pitch on every shield land.

5. Layout



All SMD parts are on L4 (bottom); the patches are on L1 (top). Each L channel sits inside an octagonal ground frame, and each S chain inside a strip frame in the axis gap between L channels. Neighbouring frames share walls, so one cover wall serves both cavities. The TA1442A and C2327J5003AHF land patterns follow the vendor drawings (Tai-Saw Rev 1, TTM Rev K). Ground pads carry filled vias (IPC-4761 VII) for the lowest ground inductance.

6. RF budget (antenna feed -> SMA, 25 C typical)

Data: project 16 Touchstone models (LNA, SAW, hybrid, diplexer), project 17 stage values for the L chain, this board's RF track lengths. Line loss is scaled with f^0.75 from 0.066 dB/cm at 2.5 GHz [ESTIMATE]. The CLM-83-2W+ loss at S is 0.30 dB [ESTIMATE]. The S LNA noise figure is 0.60 dB (NFmin 0.51 dB at 2.5 GHz + 0.1 dB match) [ESTIMATE].

GPS L1 C/A (1575.42 MHz)

#	Stage	G dB	NF dB	Cum. G dB	Cum. NF dB	Note
0	Feed line, patch pin -> hybrid	-0.013	0.013	-0.01	0.013	2.7 mm
1	HC125A hybrid (RHCP combine)	-0.361	0.361	-0.37	0.374	p16 model
2	Line, hybrid -> limiter	-0.069	0.069	-0.44	0.442	14.7 mm
3	CLM-83-2W+ limiter	-0.165	0.165	-0.61	0.607	p17
4	Line, limiter -> LNA1	-0.02	0.02	-0.63	0.627	

#	Stage	G dB	NF dB	Cum. G dB	Cum. NF dB	Note
5	LNA1 TQP3M9037 + bias	21.953	0.372	21.33	0.999	p16 model
6	B7504 #1	-0.893	0.893	20.43	1.005	
7	LNA2 TQP3M9037 + bias	21.953	0.372	42.39	1.008	
8	B7504 #2	-0.893	0.893	41.49	1.008	
9	B7504 #3	-0.893	0.893	40.6	1.008	
10	DXP.02.A combiner	-3.308	3.308	37.29	1.008	p16 model
11	Output line + DC block -> SMA	-0.175	0.175	37.12	1.008	

GPS L5 (1176.45 MHz)

#	Stage	G dB	NF dB	Cum. G dB	Cum. NF dB	Note
0	Feed line, patch pin -> hybrid	-0.027	0.027	-0.03	0.027	7.1 mm
1	HC125A hybrid (RHCP combine)	-0.191	0.191	-0.22	0.218	p16 model
2	Line, hybrid -> limiter	-0.017	0.017	-0.23	0.234	4.5 mm
3	CLM-83-2W+ limiter	-0.138	0.138	-0.37	0.372	p17
4	Line, limiter -> LNA1	-0.016	0.016	-0.39	0.389	
5	LNA1 TQP3M9037 + bias	23.606	0.394	23.22	0.783	p16 model
6	B8889 #1	-0.911	0.911	22.31	0.787	
7	LNA2 TQP3M9037 + bias	23.606	0.394	45.91	0.789	
8	B8889 #2	-0.911	0.911	45.0	0.789	
9	B8889 #3	-0.911	0.911	44.09	0.789	
10	DXP.02.A combiner	-3.138	3.138	40.95	0.789	p16 model
11	Output line + DC block -> SMA	-0.132	0.132	40.82	0.789	

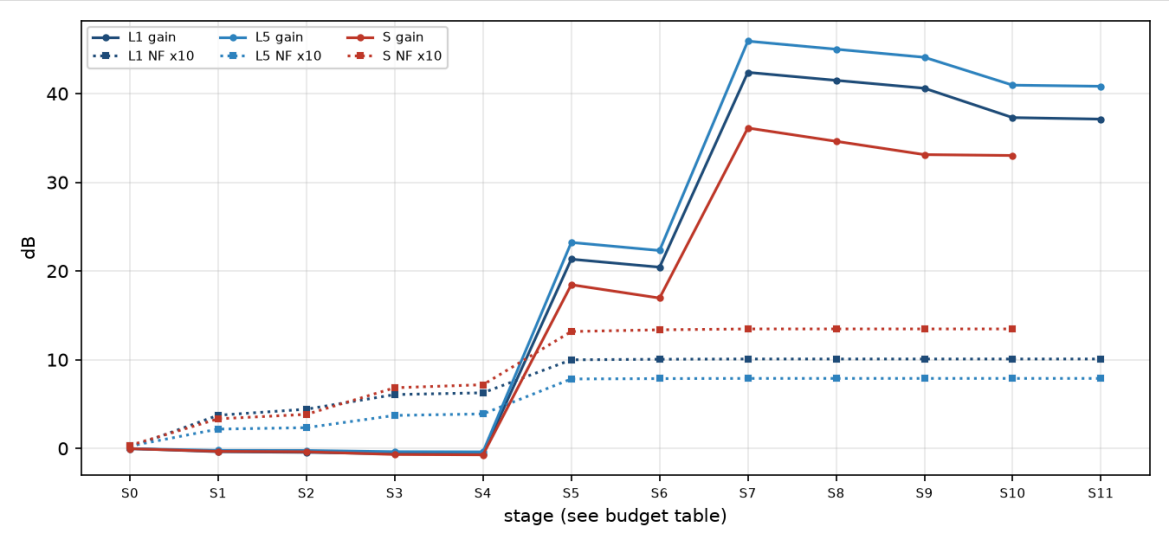
GPS L2C (1227.60 MHz)

#	Stage	G dB	NF dB	Cum. G dB	Cum. NF dB	Note
0	Feed line, patch pin -> hybrid	-0.027	0.027	-0.03	0.027	7.1 mm
1	HC125A hybrid (RHCP combine)	-0.217	0.217	-0.24	0.244	p16 model
2	Line, hybrid -> limiter	-0.017	0.017	-0.26	0.262	4.5 mm
3	CLM-83-2W+ limiter	-0.142	0.142	-0.4	0.404	p17
4	Line, limiter -> LNA1	-0.017	0.017	-0.42	0.42	
5	LNA1 TQP3M9037 + bias	23.386	0.396	22.97	0.816	p16 model
6	B8889 #1	-1.455	1.455	21.51	0.824	
7	LNA2 TQP3M9037 + bias	23.386	0.396	44.9	0.826	
8	B8889 #2	-1.455	1.455	43.44	0.826	
9	B8889 #3	-1.455	1.455	41.99	0.826	
10	DXP.02.A combiner	-3.555	3.555	38.43	0.827	p16 model
11	Output line + DC block -> SMA	-0.137	0.137	38.29	0.827	

NavIC S (2492.028 MHz)

#	Stage	G dB	NF dB	Cum. G dB	Cum. NF dB	Note
0	Feed line, patch pin -> hybrid	-0.034	0.034	-0.03	0.034	5.2 mm
1	C2327J5003AHF hybrid (RHCP combine)	-0.3	0.3	-0.33	0.334	TTM Rev K typ excess IL
2	Line, hybrid -> limiter	-0.05	0.05	-0.38	0.384	7.6 mm
3	CLM-83-2W+ limiter	-0.3	0.3	-0.68	0.684	[ESTIMATE] 0.5 dB typ broadband, 0.17 dB at L1 (p17)

#	Stage	G dB	NF dB	Cum. G dB	Cum. NF dB	Note
4	Line, limiter -> LNA1	-0.034	0.034	-0.72	0.718	
5	LNA1 TQP3M9037 (2.3-2.6 GHz match)	19.168	0.6	18.45	1.318	NFmin 0.51 dB @2.5 GHz + 0.1 match [ESTIMATE]
6	TA1442A #1	-1.5	1.5	16.95	1.337	Tai-Saw model (p16 A24)
7	LNA2 TQP3M9037	19.168	0.6	36.12	1.347	
8	TA1442A #2	-1.5	1.5	34.62	1.347	
9	TA1442A #3	-1.5	1.5	33.12	1.347	
10	Output line + DC block -> SMA	-0.095	0.095	33.02	1.347	incl. SMA launch [ESTIMATE]



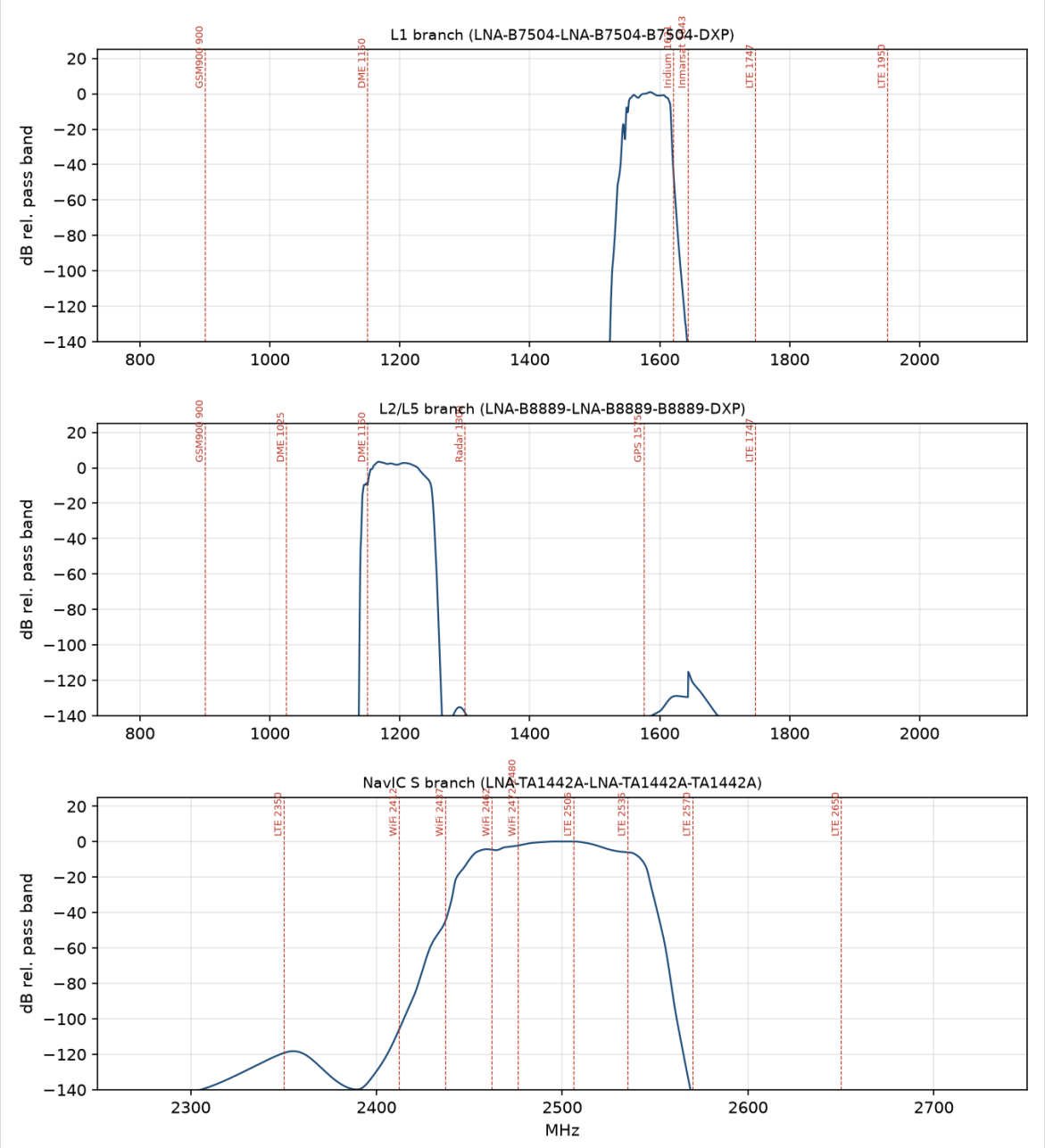
The system noise figure is set ahead of LNA1: hybrid, limiter and lines add 0.4-0.7 dB. Gain at the SMA is 33-41 dB, before the cable to the CRPA electronics.

7. Interference rejection, linearity and filter power

7.1 Rejection

Branch	Interferer	MHz	Rejection dB (rel. pass band)
L1	GSM900 / LTE B8 UL 900	900	>80
L1	DME / TACAN 1150	1150	>80
L1	Inmarsat UL 1643	1643	>80
L1	Iridium 1621	1621	43.3
L1	LTE B3 UL 1747	1747	>80
L1	LTE B1 UL 1950	1950	>80
L2/L5	GSM900 / LTE B8 UL 900	900	>80
L2/L5	DME / TACAN 1025	1025	>80
L2/L5	DME / TACAN 1150	1150	9.9
L2/L5	Radar / ATC 1300	1300	>80
L2/L5	GPS L1 1575	1575	>80
L2/L5	LTE B3 UL 1747	1747	>80
S	LTE B40 (2300-2400) 2350	2350	>80
S	WiFi ch1 / BT low 2412	2412	>80
S	WiFi ch6 2437	2437	45.0
S	WiFi ch11 2462	2462	4.4

Branch	Interferer	MHz	Rejection dB (rel. pass band)
S	WiFi ch13 / BT high 2472-2480	2476	2.2
S	LTE B41 low 2506	2506	-0.1
S	LTE B7 UL 2535	2535	5.9
S	LTE B41 / B7 2570	2570	>80
S	LTE B41 2650	2650	>80



The cascaded model gives over 100 dB far from the bands. A real board leaks around the filters (cover cavities, ground, bias lines), so anything above 80 dB is quoted as ">80". Near-band results that matter: - **NavIC S:** the TA1442A curve stays within about 3 dB loss from roughly 2445 to 2540 MHz. The "5 MHz bandwidth" in its datasheet is only the window it is specified over. WiFi channels 11-13, Bluetooth, LTE B41 (2496-2690, used in India) and B7 uplink (2500-2570) therefore pass almost unattenuated. Wi-Fi ch6 gets 45 dB. - **L2/L5:** DME / TACAN replies up to 1150-1213 MHz overlap the wide 1164-1240 MHz band; only about 10 dB at 1150 MHz. - **L1:** Iridium (1616-1626 MHz) 43 dB; Inmarsat uplink > 80 dB.

7.2 Linearity and SAW power rating

Branch	2nd SAW	Rating	Gain ahead of SAW #1 / #2	CW at antenna that reaches the rating (SAW #1 / #2)
L1	B7504	15 dBm (out of band, 5000 h)	about 21 / 42 dB	-6.3 / -27.4 dBm
L2/L5	B8889	5 dBm CW (2000 h at 55 C)	about 23 / 45 dB	-18.2 / -40.9 dBm
S	TA1442A	10 dBm	about 17 / 35 dB	-8.4 / -26.1 dBm

The limiter caps the LNA1 input at about +11.5 dBm, but LNA2 can still deliver +20 dBm into SAW #2. A jammer 30-40 dB above the noise floor near the antenna is normal for a CRPA. The present gain distribution then overdrives SAW #2 and compresses the chain before the nulling electronics see the signal.

Rev B recommendations:

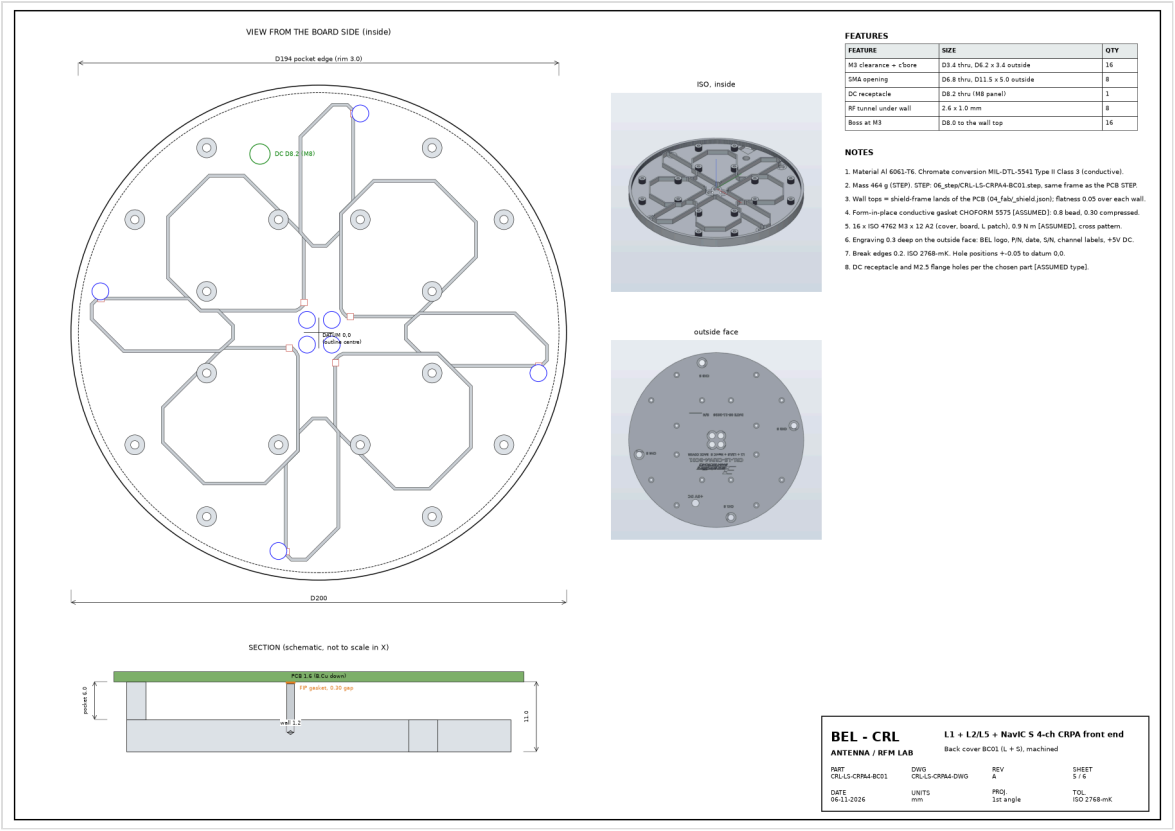
1. Cut the gain to about 25-28 dB per branch. Use one LNA ahead of the first SAW and a lower-gain second stage, or a 3-6 dB pad between LNA2 and SAW #2. This raises the input P1dB to about -10 dBm and keeps SAW #2 below rating.
2. For S, add a second-stage filter with a sharp upper skirt (ceramic / BAW, LTE B41 and B7 rejection) or a tighter S SAW. For L2/L5, decide whether the band must start at 1164 MHz (DME); 1166-1240 with a steeper filter removes DME.
3. Specify the receiver-side requirement: required gain at the antenna port, J/S to survive, recovery time.

8. Power and thermal

Item	Value
LNAs	24 x TQP3M9037, 70 mA typ / 90 mA max at 5 V
Supply current	1.68 A typ / 2.16 A max
Power	8.4 W typ / 11.3 W max (5.25 V)
LNA dissipation	0.35 W each, theta_jc 53.4 C/W -> +18.7 C over the case
Tj at 85 C case	103.7 C (limit for > 1e6 h MTTF: 190 C)
Heat path	exposed pad -> filled vias -> L3 / L2 GND -> cover walls through the conductive gasket

The thermal margin is large. The open point is the case temperature inside the sealed cover at the platform's maximum ambient (section 9).

9. Back cover BC01 (L + S)



One machined Al 6061-T6 part, D200 mm, 11 mm high, 464 g, chromate conversion (conductive). Walls 1.2 mm wide run on every shield-frame land, so each L channel and each S chain has its own cavity. A form-in-place conductive gasket goes on the wall tops. RF tunnels 2.6 x 1.0 mm pass under the walls. There are 8 recessed SMA openings, an M8 DC receptacle, and 16 x M3 through cover, board and L patch. The outside face is engraved with the BEL logo, P/N, date, S/N and channel labels.

10. Manufacturing package

File	Content
`04_fab/CRL-LS-CRPA4_Gerbers.zip`	Gerber X2 (4 Cu, mask, paste, silk, outline), Excellon PTH/NPTH + drill map
`04_fab/CRL-LS-CRPA4_CPL_bottom.csv`	pick-and-place, bottom side
`04_fab/CRL-LS-CRPA4.d356`	IPC-D-356 netlist for the bare-board electrical test
`04_fab/CRL-LS-CRPA4_BOM.csv / .xlsx`	40 lines incl. mechanical items
`04_fab/CRL-LS-CRPA4_FAB_NOTES.md`	stack-up, impedance, finish, via plugging, assembly and test
`06_step/CRL-LS-CRPA4_PCB.step`, `06_step/CRL-LS-CRPA4-BC01.step`	board and cover in one frame
`07_drawings/CRL-LS-CRPA4_DRAWINGS.pdf`	6 sheets: fab, array, assembly bottom / top, cover, block diagram

11. Open items and review gaps

- Inputs needed: CRL-LS-SP01 drawing and measured S11 / axial ratio; L patch fixing holes (board D3.6 / D1.35 vs GPDF6010.A D1.8 / D1.1); supply bus (5 V or 28 V); platform (MIL-STD-810 / 461 levels); CRPA processor gain / match / delay requirements; radome; interference environment.
- Not yet built: test boards (one L channel, one S chain, THRU line) and their cover.

- Full list: `05_docs/CRL-LS-CRPA4_design_review_gaps.md`.

12. Assumptions

Values marked [ESTIMATE] / [ASSUMED] in the tables; project 16 `assumptions_log.md` (copied in `00_reference/p16`) for the component models. All data at 25 C typical; no temperature, part-to-part or ageing spread.